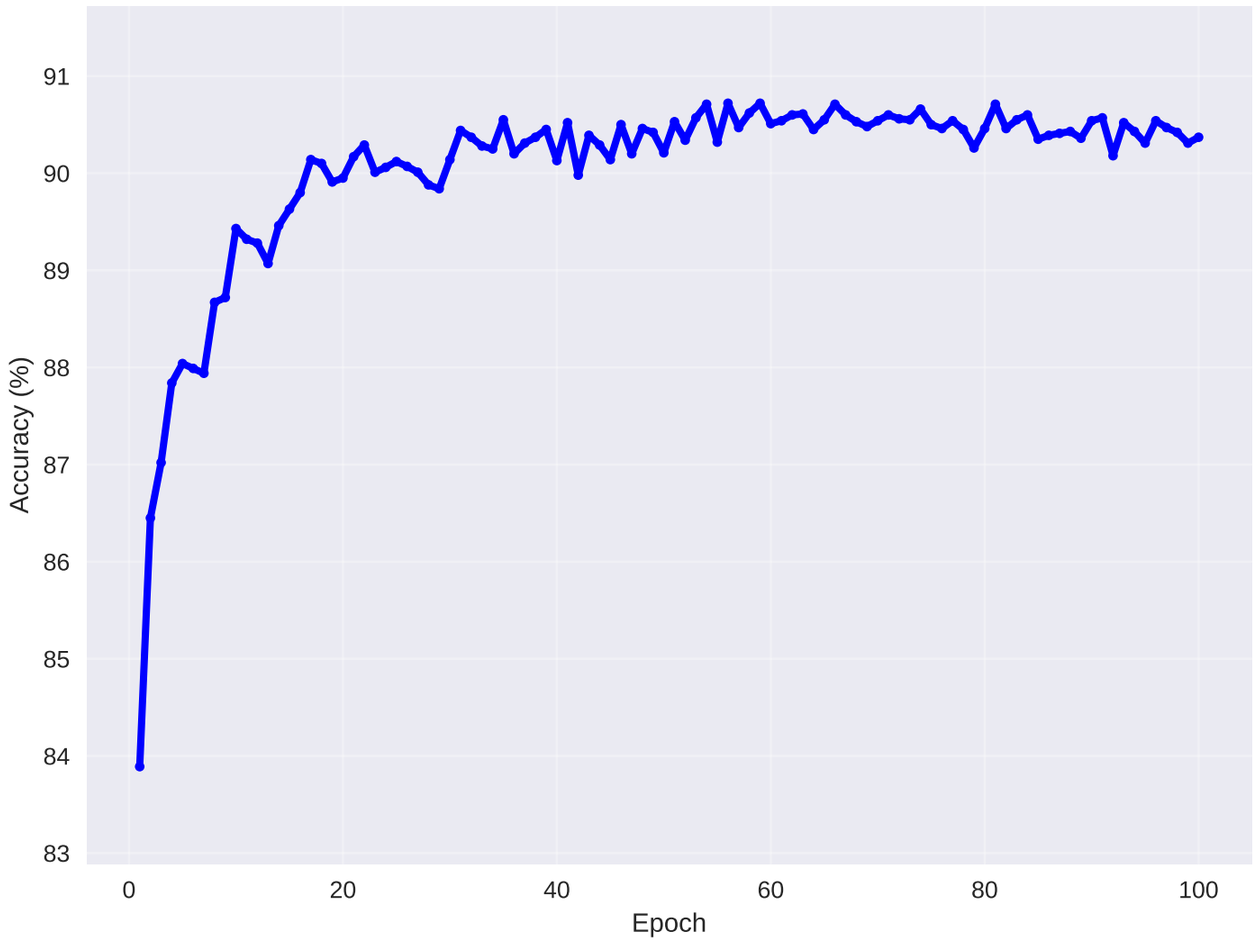
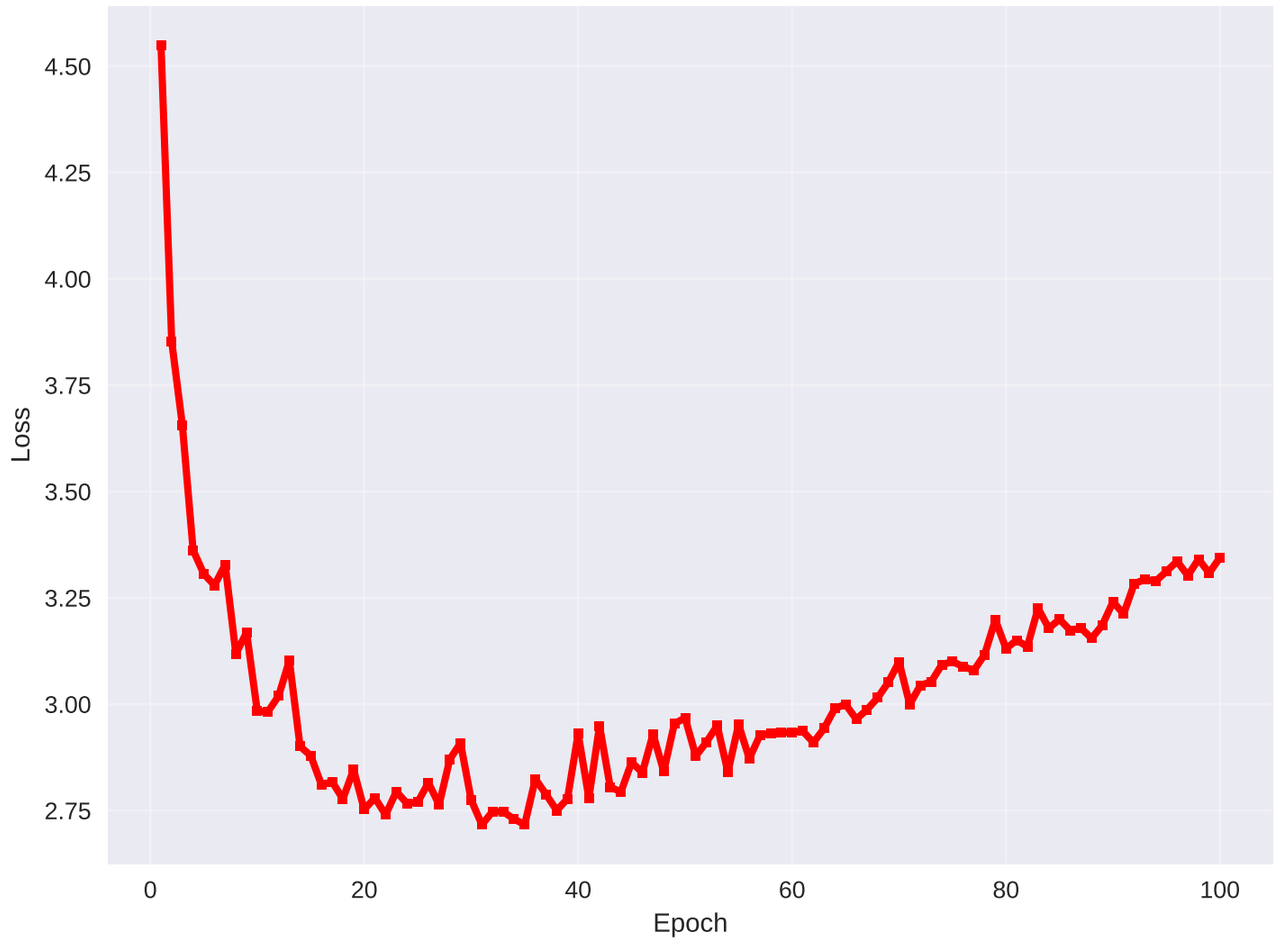
Federated Learning - Main Metrics Overview

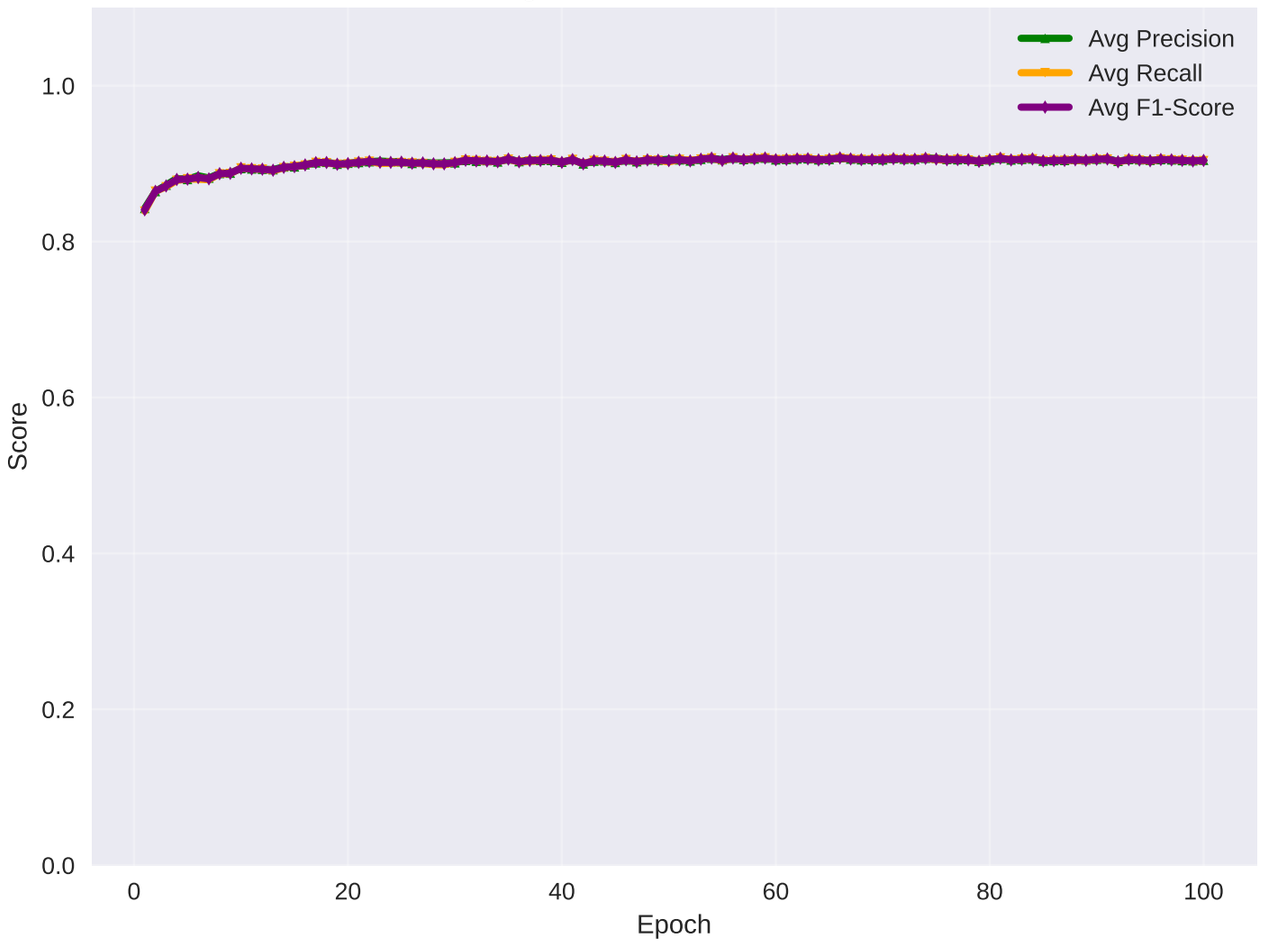
□ Global Model Accuracy Evolution



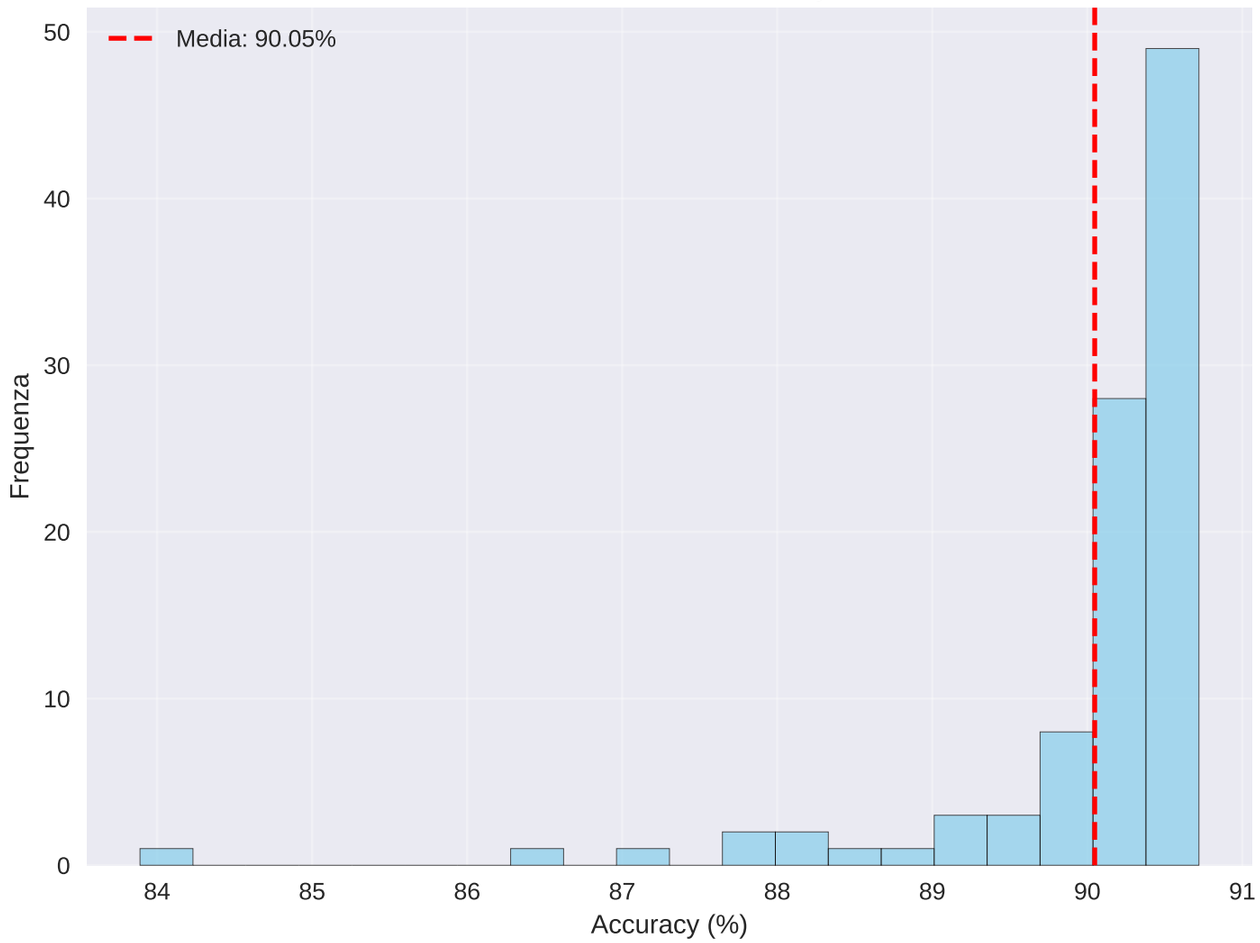
□ Global Model Loss Evolution



□ Average Precision, Recall & F1-Score

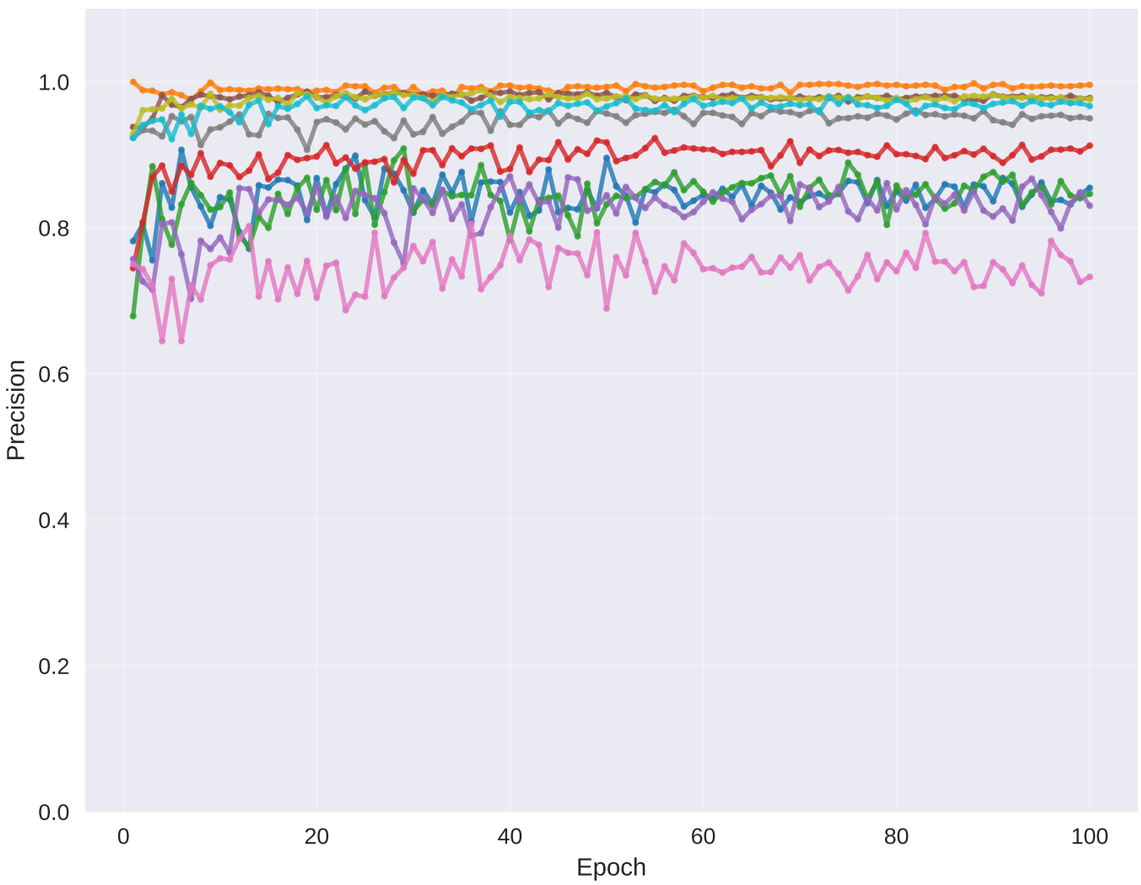


□ Distribuzione Accuracy

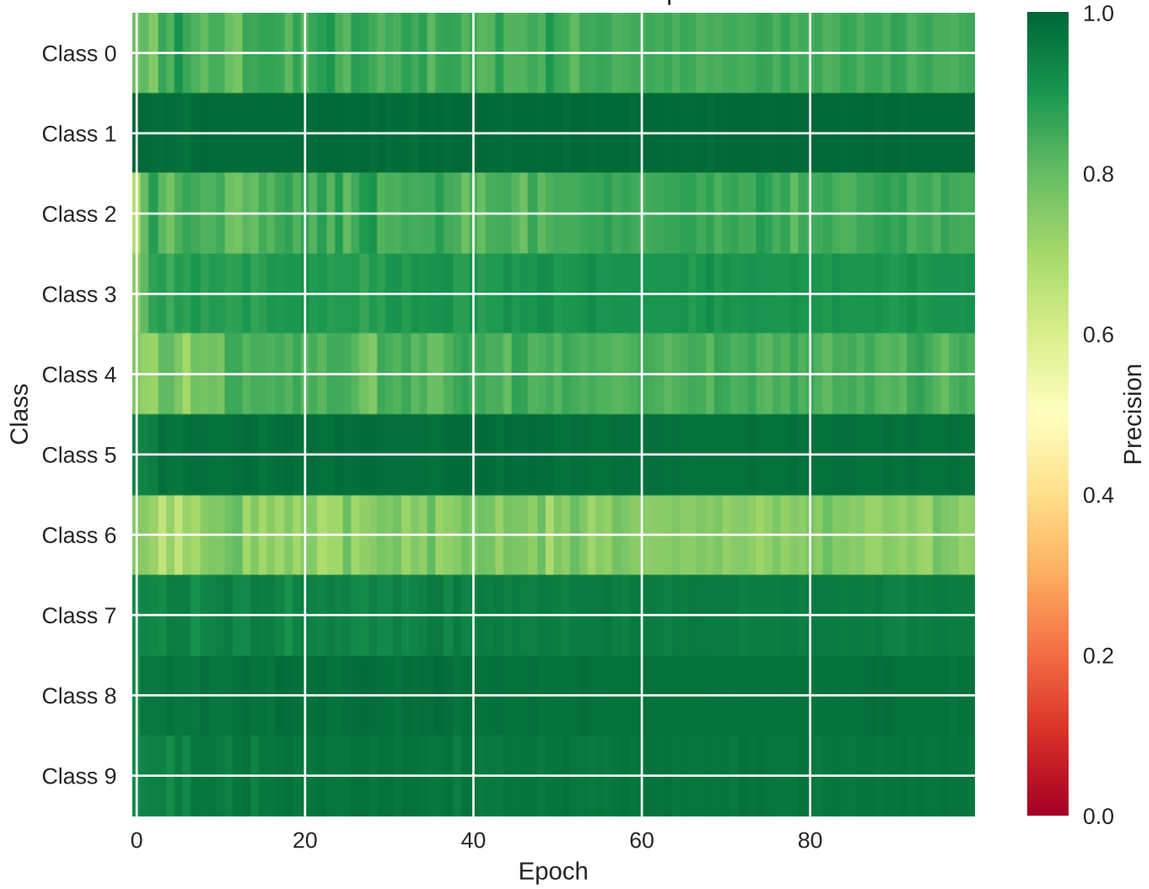


Per-Class Precision Analysis

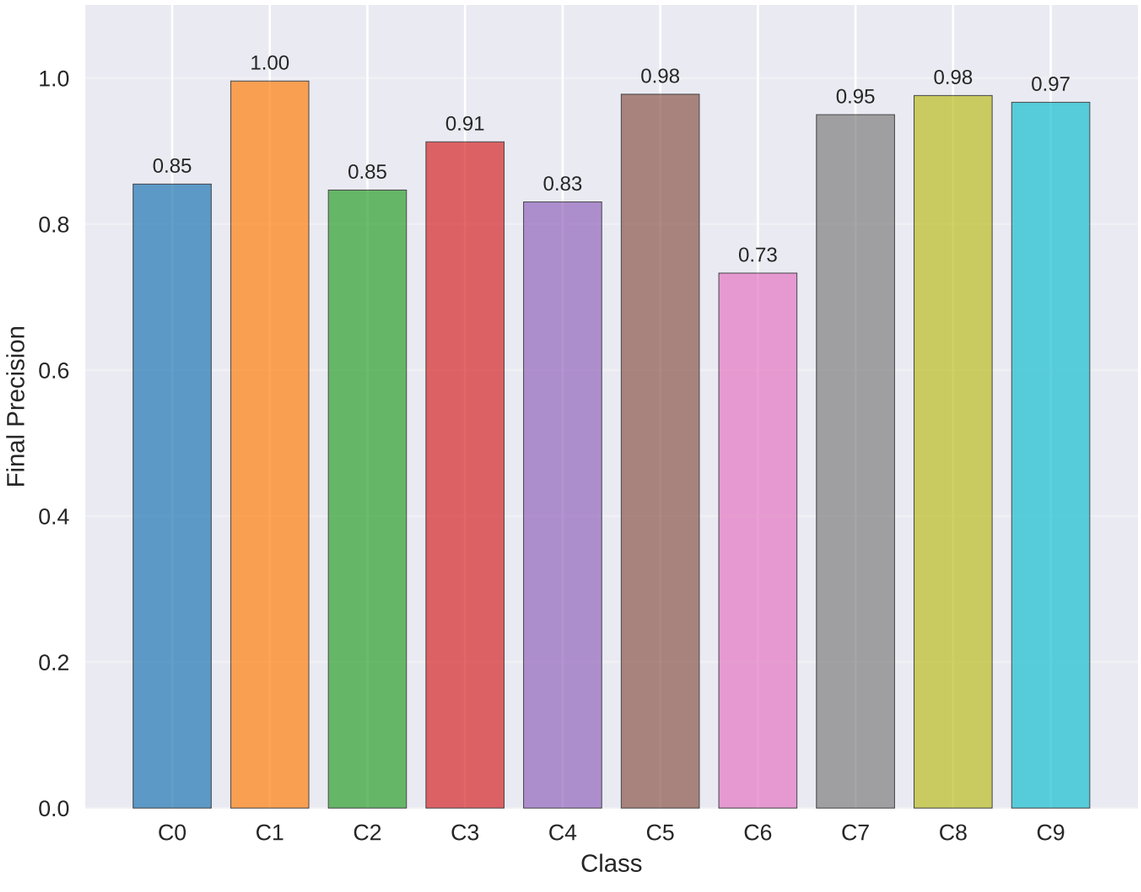
□ Per-Class Precision Evolution



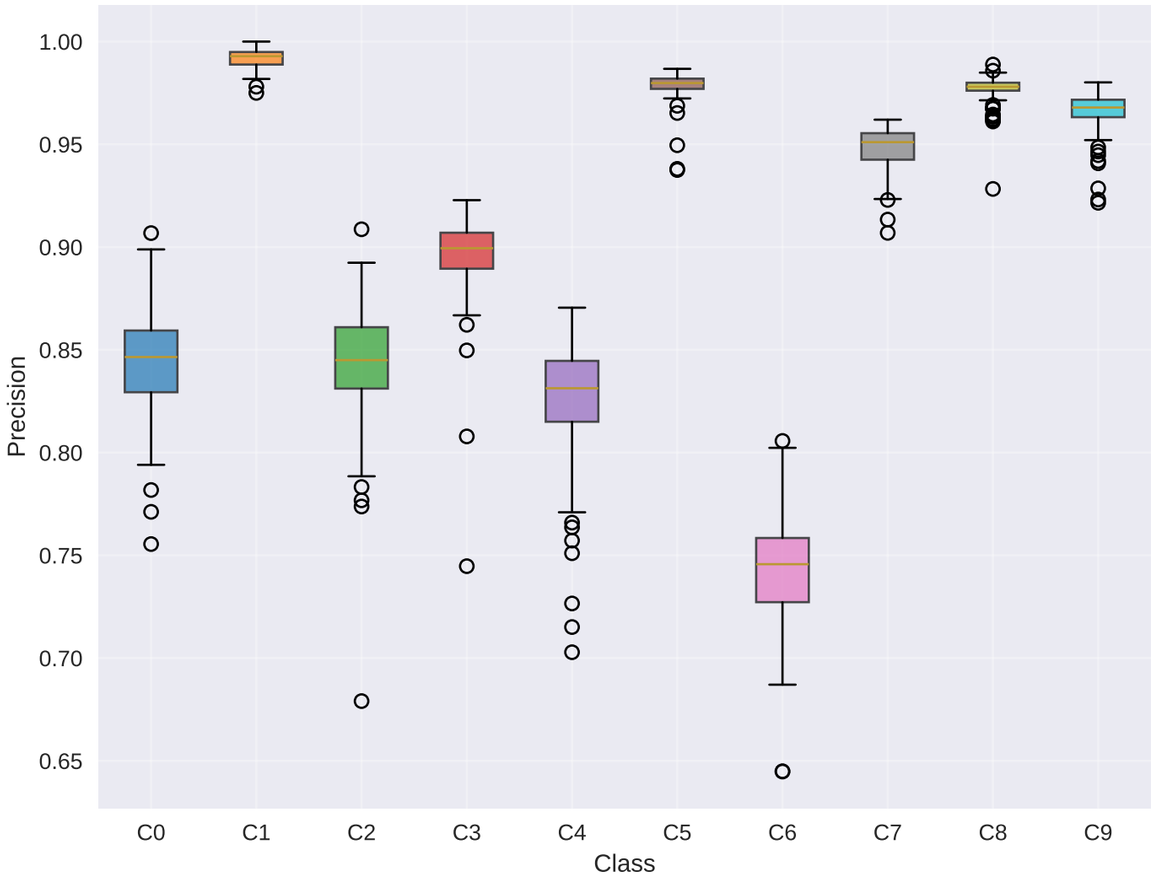
□ Precision Heatmap



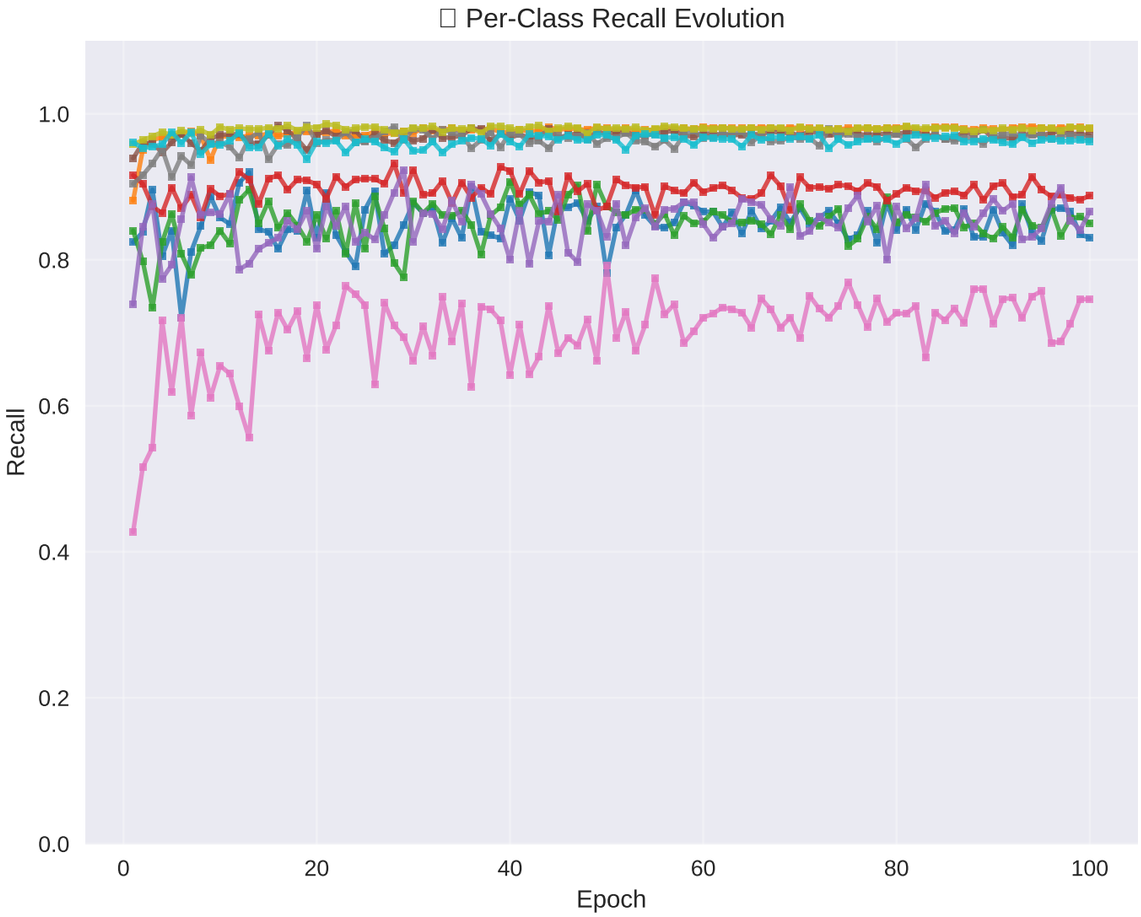
□ Final Precision per Class

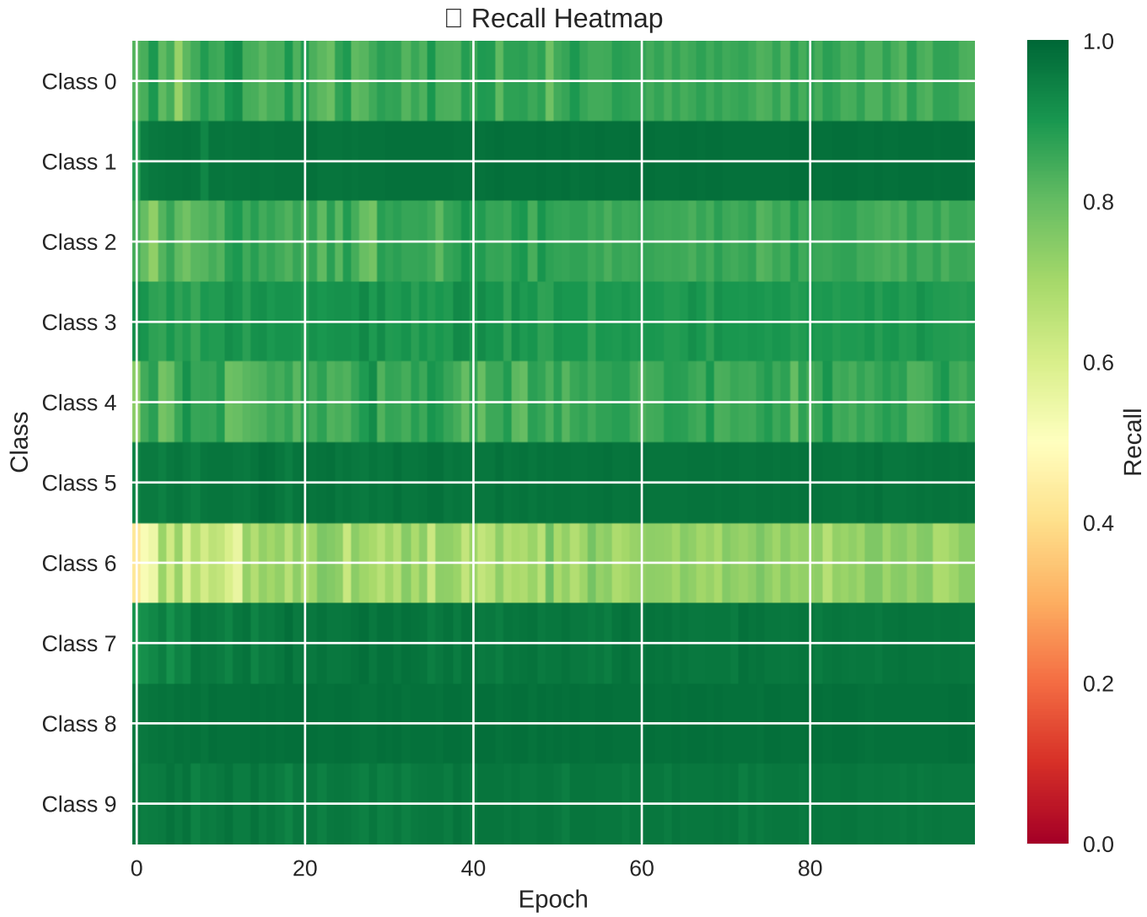


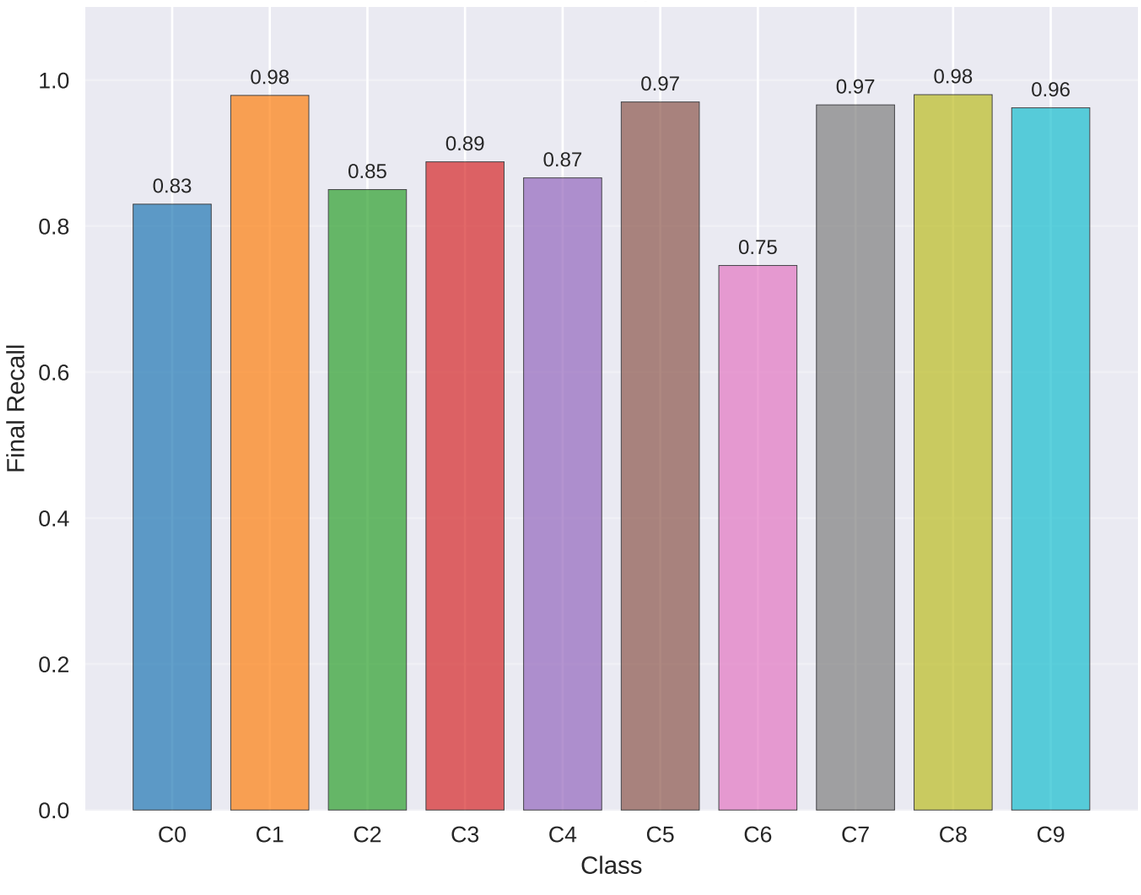
□ Precision Distribution per Class

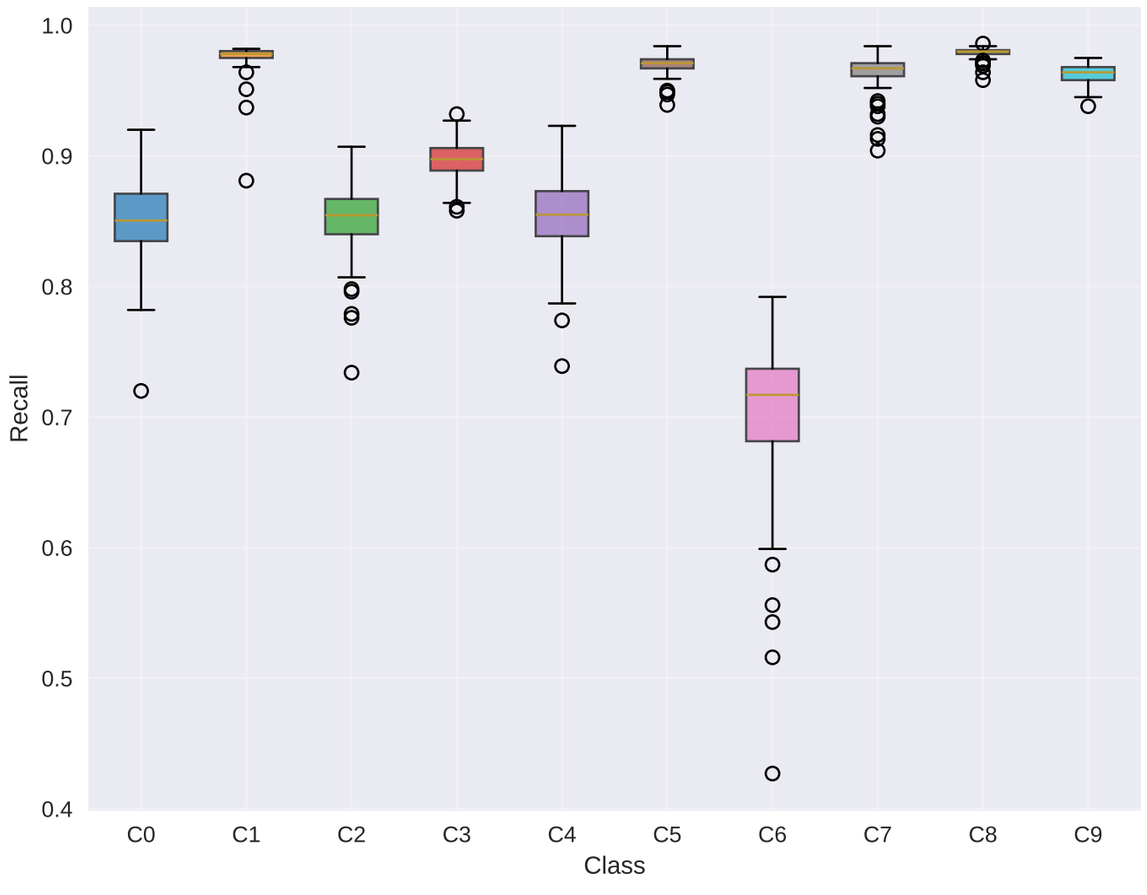


Per-Class Recall Analysis

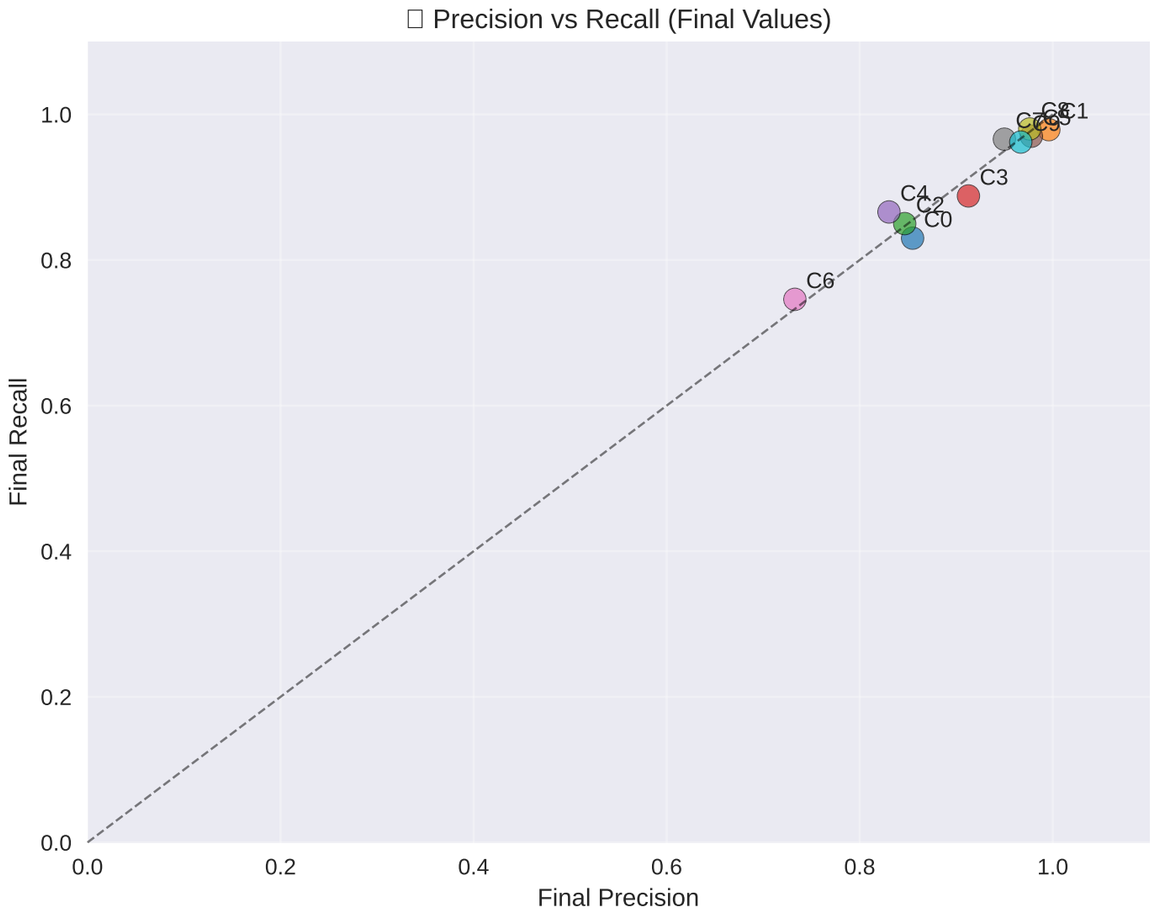




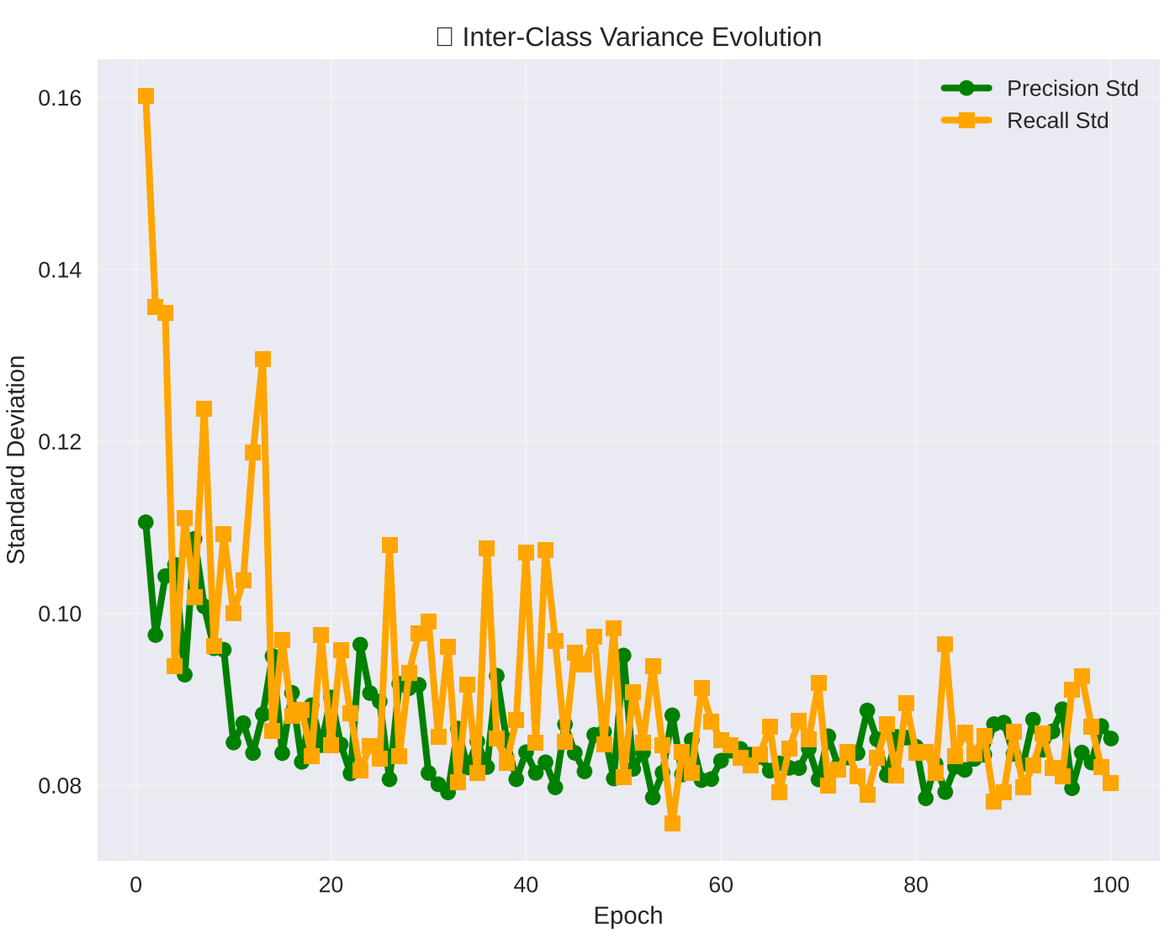


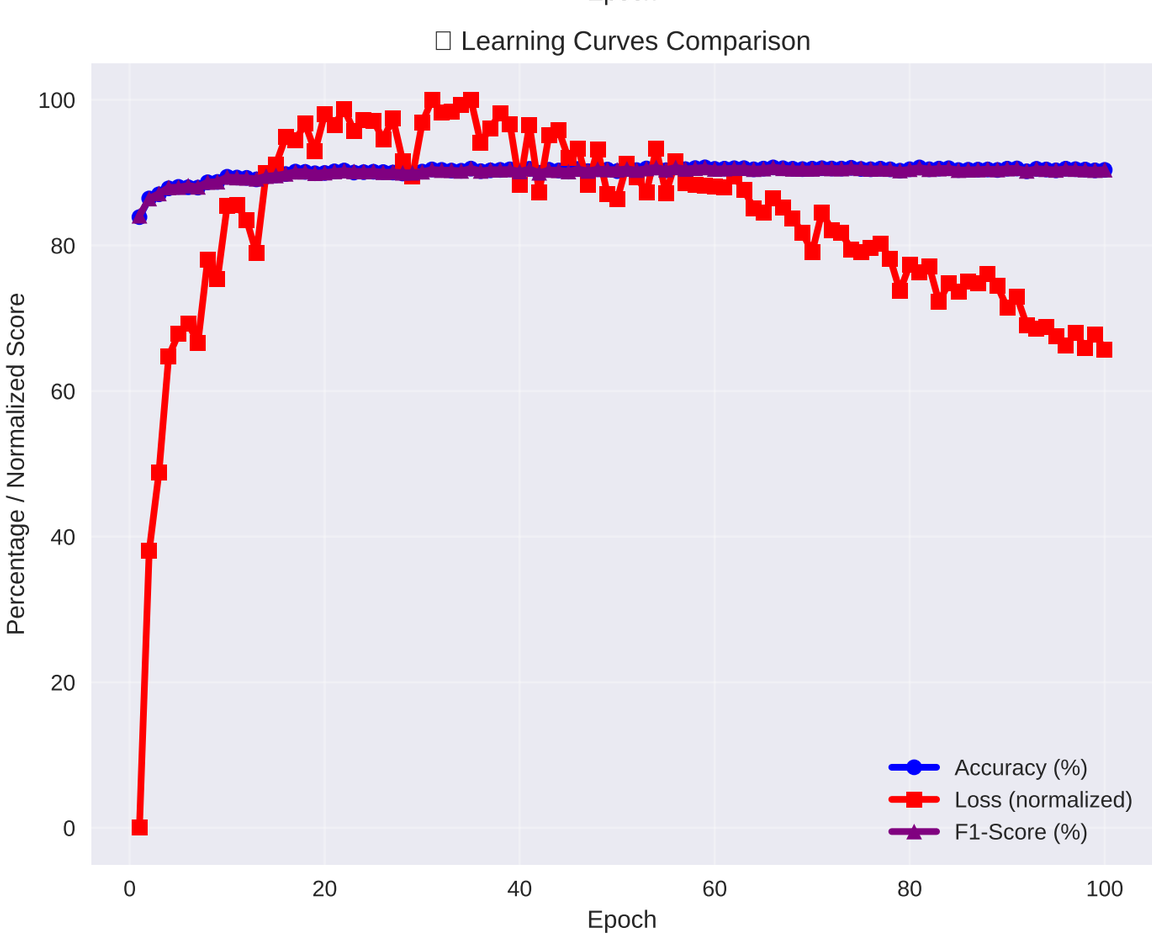


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-05 14:53:57
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 90.37%
• Loss: 3.3446
• Average Precision: 0.904
• Average Recall: 0.904
• Average F1-Score: 0.904

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.48%
• Riduzione Loss: +1.2049
• Stabilità Accuracy (CV): 0.0110

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.996)
• Peggiore Precision: Classe 6 (0.733)
• Migliore Recall: Classe 8 (0.980)
• Peggiore Recall: Classe 6 (0.746)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.855, R=0.830, F1=0.842
• Classe 1: P=0.996, R=0.979, F1=0.987
• Classe 2: P=0.847, R=0.850, F1=0.848
• Classe 3: P=0.913, R=0.888, F1=0.900
• Classe 4: P=0.830, R=0.866, F1=0.848
• Classe 5: P=0.978, R=0.970, F1=0.974
• Classe 6: P=0.733, R=0.746, F1=0.739
• Classe 7: P=0.950, R=0.966, F1=0.958
• Classe 8: P=0.976, R=0.980, F1=0.978
• Classe 9: P=0.967, R=0.962, F1=0.964

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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